Coloured Noise Circuit (CNC)

Although a noise filter is already integrated into the FORMANT's NOISE module, its parameters cannot be adjusted from the outside (i.e., via the front panel). For those who wish to vary the FORMANT's noise characteristics across the full range of nuances—and who do not shy away from mechanically modifying the front panel—a CNC machine provides the ideal tool for the job.

The circuit for the adjustable noise filter is shown in Figure 1. It consists of a tone control circuit—specifically modified for this application—configured as a negative feedback network.

Construction and Assembly

Figure 2 shows the PCB layout and the corresponding component placement plan. If the E12-series values for capacitors C3, C4, and C5 are difficult to obtain, they can be realized by connecting smaller values in parallel (e.g., C4 = 4.7 nF + 1.0 nF; C3, C5 = 4.7 nF + 3.3 nF). The resulting mismatch of 0.1 nF or 0.2 nF is negligible, as it falls within the tolerance range of most film capacitors (10%).

The PCB connections for P1 and P2 are designed to accommodate the soldering of terminal pins for rotary potentiometers.

In the version using rotary potentiometers, the "coloration" can also be adjusted from the outside. However, this requires drilling the appropriate holes in the NOISE front panel. To avoid damaging the front panel, the "drilling guidelines" (see Chapter 1, Portamento Switch) should be observed. A design for a front panel label is shown in Figure 3. Similar to the triangle/sine converter or the digital noise generator, the CNC PCB can be mounted to the noise PCB using 20–30 mm spacers. Consequently, the "colored noise" circuitry surrounding IC2 on the FORMANT noise module can be omitted. In this case, the CNC input is connected to the junction point of Pin 6 (IC1), C4, C5, and R5, and the CNC output is connected to the junction point of Pin 6 (IC2), C6, R13, R14, and R16.

Calibration

Calibrating the Coloured Noise Circuit presents no difficulties. The wiper of P3 is adjusted so that the output voltage matches that of the white noise output. A standard multimeter (with an integrating display) is sufficient for this measurement.

No definitive guidelines can be given for the settings of P1 and P2. The adjustment is largely a matter of personal preference and varies from person to person.

Applications

In the FORMANT system, the Coloured Noise Circuit is primarily used to generate specific timbres—that is, frequency distributions of the noise signal—without the need for a VCF. This circuit is particularly well-suited for special effects such as gale-force winds, jet engines, the sound of crashing surf, and so on. Furthermore, any other sound structures to which coloured noise is added gain an expanded range of possibilities for variation and imitation.

Parts list for Figure 1

Resistors (carbon film, 5%):

R1, R2, R3, R7 = 2.7k
R4 = 3.3k
R5 = 68k
R6 = 820R

Potentiometers:

P1, P2 = 100k lin.
P3 = 4.7k (trimmer)

Capacitors:

C1 = 2.2µF/25V
C2 = 220nF
C3, C5 = 8.2nF (see text)
C4 = 5.6nF (see text)

Semiconductors:

T1 = TUN
IC1 = µA741C (Mini-DIP)

Miscellaneous:

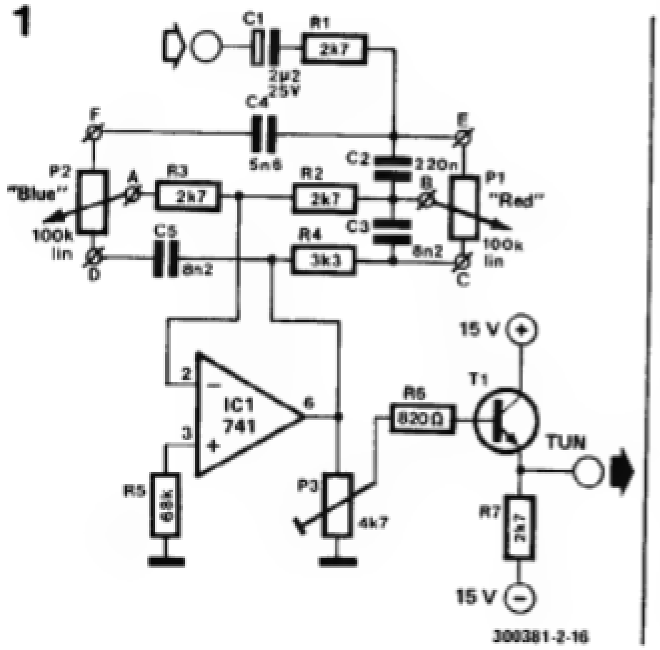
2 spacer sleeves

Figures

- Figure 1. Schematic diagram for the "coloured noise" circuit. In principle, this is a specially designed tone control.
- Figure 2. PCB layout and component placement plan for the circuit shown in Figure 1.

- Figure 3. Suggested front panel design for the expanded NOISE module. The additional labels signify: RED C.N. (red coloured noise) = red-tinted noise; BLUE C.N. (blue coloured noise) = blue-tinted noise.

1



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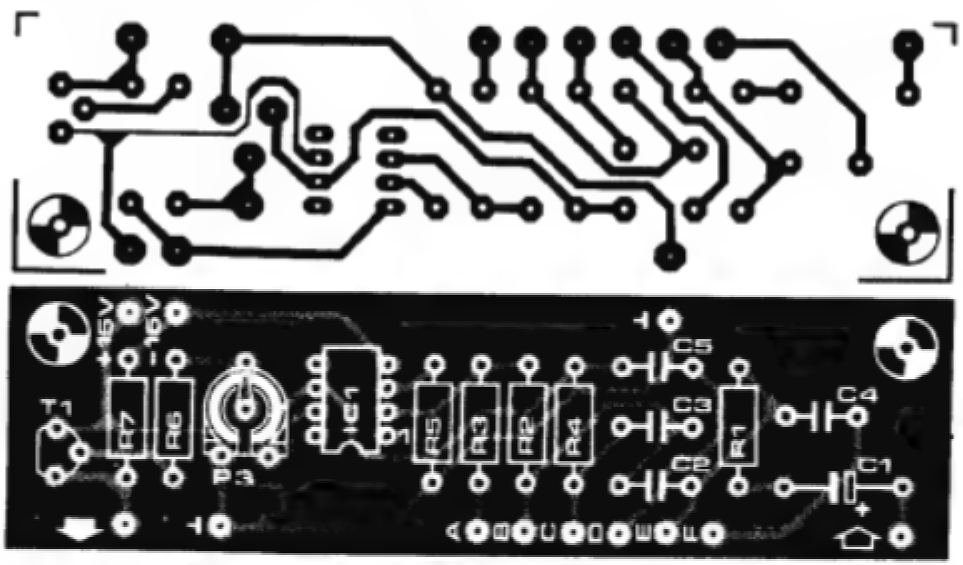

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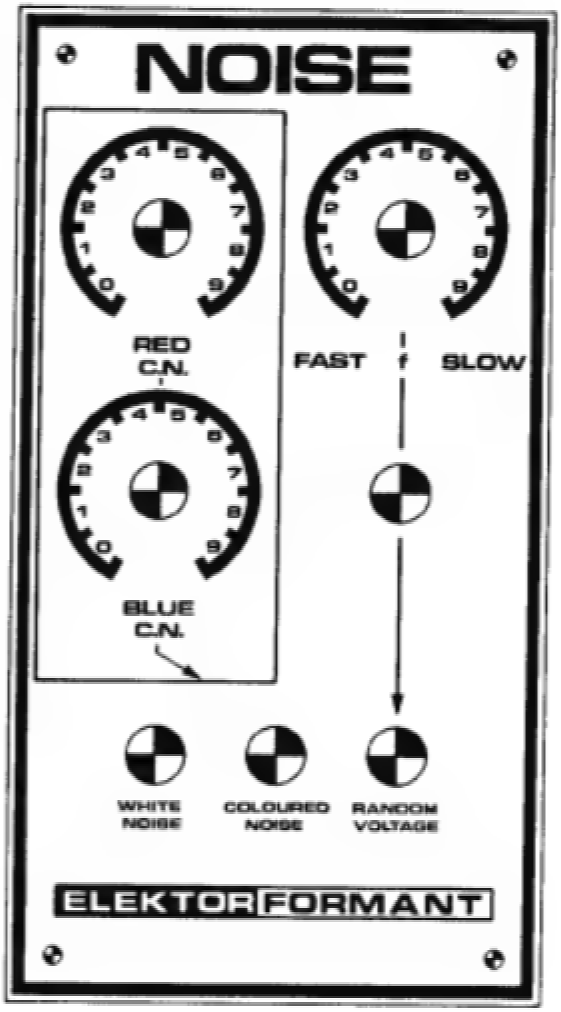
NOISE
0 1 2 3 4 5 6 7 8 9
RED
C.N.
0 1 2 3 4 5 6 7 8 9
BLUE
C.N.
0 1 2 3 4 5 6 7 8 9
FAST
f
SLOW
WHITE
NOISE
COLOURED
NOISE
RANDOM
VOLTAGE
ELEKTOR FORMANT